

1. Inverted Right Triangle Pattern

Program

```
#include <iostream>

using namespace std;

int main() {

    int n = 5;

    for(int i = n; i >= 1; i--) {

        for(int j = 1; j <= i; j++) {

            cout << "* ";

        }

        cout << endl;

    }

    return 0;

}
```

Output

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int n = 5;
5     for(int i = n; i >= 1; i--) {
6         for(int j = 1; j <= i; j++) {
7             cout << "* ";
8         }
9         cout << endl;
10    }
11    return 0;
12 }
```

OUTPUT

```
* * * * *
* * * *
* * *
* *
*
```

2. Check Whether a Number is a Strong Number

Program

```
#include <iostream>

using namespace std;

int main() {

    int num = 145;

    int temp = num;

    int sum = 0;

    while(temp > 0) {

        int digit = temp % 10;

        int fact = 1;

        for(int i = 1; i <= digit; i++)

            fact = fact * i;

        sum = sum + fact;

        temp = temp / 10;

    }

    if(sum == num)

        cout << num << " is a Strong Number";

    else

        cout << num << " is not a Strong Number";

    return 0;

}
```

Output

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int num = 145;
5     int temp = num;
6     int sum = 0;
7     while(temp > 0) {
8         int digit = temp % 10;
9         int fact = 1;
10        for(int i = 1; i <= digit; i++)
11            fact = fact * i;
12        sum = sum + fact;
13        temp = temp / 10;
14    }
15    if(sum == num)
16        cout << num << " is a Strong Number";
17    else
18        cout << num << " is not a Strong Number";
19    return 0;
20 }
```

OUTPUT

```
145 is a Strong Number
```

3. Count the Frequency of Each Digit in a Number

Program

```
#include <iostream>

using namespace std;

int main() {

    int num = 122333;

    int a[10] = {0};

    while(num > 0) {

        int d = num % 10;

        a[d]++;

        num = num / 10;

    }

    for(int i = 0; i < 10; i++) {

        if(a[i] > 0)

            cout << i << " = " << a[i] << endl;
```

```

    }

    return 0;
}

```

Output

```

1 #include <iostream>
2 using namespace std;
3 int main() {
4     int num = 122333;
5     int a[10] = {0};
6     while(num > 0) {
7         int d = num % 10;
8         a[d]++;
9         num = num / 10;
10    }
11    for(int i = 0; i < 10; i++) {
12        if(a[i] > 0)
13            cout << i << " = " << a[i] << endl;
14    }
15    return 0;
16 }
17

```

OUTPUT

```

1 = 1
2 = 2
3 = 3

```

4. Count the Number of Divisors of a Number

Program

```

#include <iostream>

using namespace std;

int main() {

    int num = 12;

    int count = 0;

    for(int i = 1; i <= num; i++) {

        if(num % i == 0)

```

```

        count++;
    }

    cout << "Number of Divisors = " << count;

    return 0;
}

```

Output

```

1 #include <iostream>
2 using namespace std;
3 int main() {
4     int num = 12;
5     int count = 0;
6     for(int i = 1; i <= num; i++) {
7         if(num % i == 0)
8             count++;
9     }
10    cout << "Number of Divisors = " << count;
11    return 0;
12 }
13

```

OUTPUT

```

Number of Divisors = 6

```

5. Calculate Compound Interest

Program

```

#include <iostream>

#include <cmath>

using namespace std;

int main() {

    float p = 1000;

    float r = 10;

    float t = 2

```

```
float amount = p * pow(1 + r / 100, t);  
  
float ci = amount - p;  
  
cout << "Compound Interest = " << ci << endl;  
  
cout << "Total Amount = " << amount;  
  
return 0;  
  
}
```

Output

```
1 #include <iostream>  
2 #include <cmath>  
3 using namespace std;  
4 int main() {  
5     float p = 1000;  
6     float r = 10;  
7     float t = 2;  
8     float amount = p * pow(1 + r / 100, t);  
9     float ci = amount - p;  
10    cout << "Compound Interest = " << ci << endl;  
11    cout << "Total Amount = " << amount;  
12    return 0;  
13 }  
14
```

OUTPUT

```
Compound Interest = 210  
Total Amount = 1210
```